# VANISHING, SUPPORT AND (S_2) PROPERTIES OF THE DUALIZING MODULE OVER A LOCAL BASE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0AWN (sources/stacks-duality/pinned-original-source.tex line 6125).

*Allowed standard prerequisites (editorial).*

- The dimension function of the normalised dualizing complex. Same chapter, Lemma labelled lemma-good-dualizing-dimension-function, cited at proof lines 6156-6160. Identifies the dimension function with the dimension of the closure at closed points.
- Local vanishing and depth statements for a dualizing complex. Dualizing Complexes, Lemmas labelled dualizing-lemma-sitting-in-degrees and dualizing-lemma-depth-dualizing-module, cited at proof lines 6166-6170. Give the four properties after localising.
- The dualizing module of a proper scheme over a local base. Same chapter, Lemma labelled lemma-dualizing-module, cited at proof lines 6170-6172. Alternative deduction of the last two statements.

*Source reference.* Original section 0AWI: sources/stacks-duality/pinned-original-source.tex, source lines 6070–6070. Same-chapter pointer into the author environment example at source line 6070; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWM: sources/stacks-duality/pinned-original-source.tex, source lines 5856–5856. Same-chapter pointer into the author environment lemma at source line 5856; invoked at proof line 6141 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0A7U: sources/stacks-chow/tags.txt, source lines 12892–12892. Cross-chapter standard prerequisite located in the project tag list

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staged in the local snapshot; invoked at proof line 6149 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWE: sources/stacks-chow/tags.txt, source lines 13717–13717. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6150 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0AWK: sources/stacks-duality/pinned-original-source.tex, source lines 6101–6101. Same-chapter pointer into the author environment lemma at source line 6101; invoked at proof line 6153 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $X/A$ with $\omega_X^\bullet$ and $\omega_X$ be as in Example 0AWI. Then*

(1) $H^i(\omega_X^\bullet) \neq 0 \Rightarrow i \in \{-\dim(X), \ldots, 0\}$,
(2) *the dimension of the support of $H^i(\omega_X^\bullet)$ is at most $-i$,*
(3) *$\mathit{Supp}(\omega_X)$ is the union of the components of dimension $\dim(X)$, and*
(4) *$\omega_X$ has property $(S_2)$.*

*Proof.* Let $\delta_X$ and $\delta_S$ be the dimension functions associated to $\omega_X^\bullet$ and $\omega_S^\bullet$ as in Lemma 0AWM. As $X$ is proper over $A$, every closed subscheme of $X$ contains a closed point $x$ which maps to the closed point $s \in S$ and $\delta_X(x) = \delta_S(s) = 0$. Hence $\delta_X(\xi) = \dim(\overline{\{\xi\}})$ for any point $\xi \in X$. Hence we can check each of the statements of the lemma by looking at what happens over $\mathrm{Spec}(\mathcal{O}_{X,x})$ in which case the result follows from Dualizing Complexes, Lemmas 0A7U and 0AWE. Some details omitted. The last two statements can also be deduced from Lemma 0AWK. □